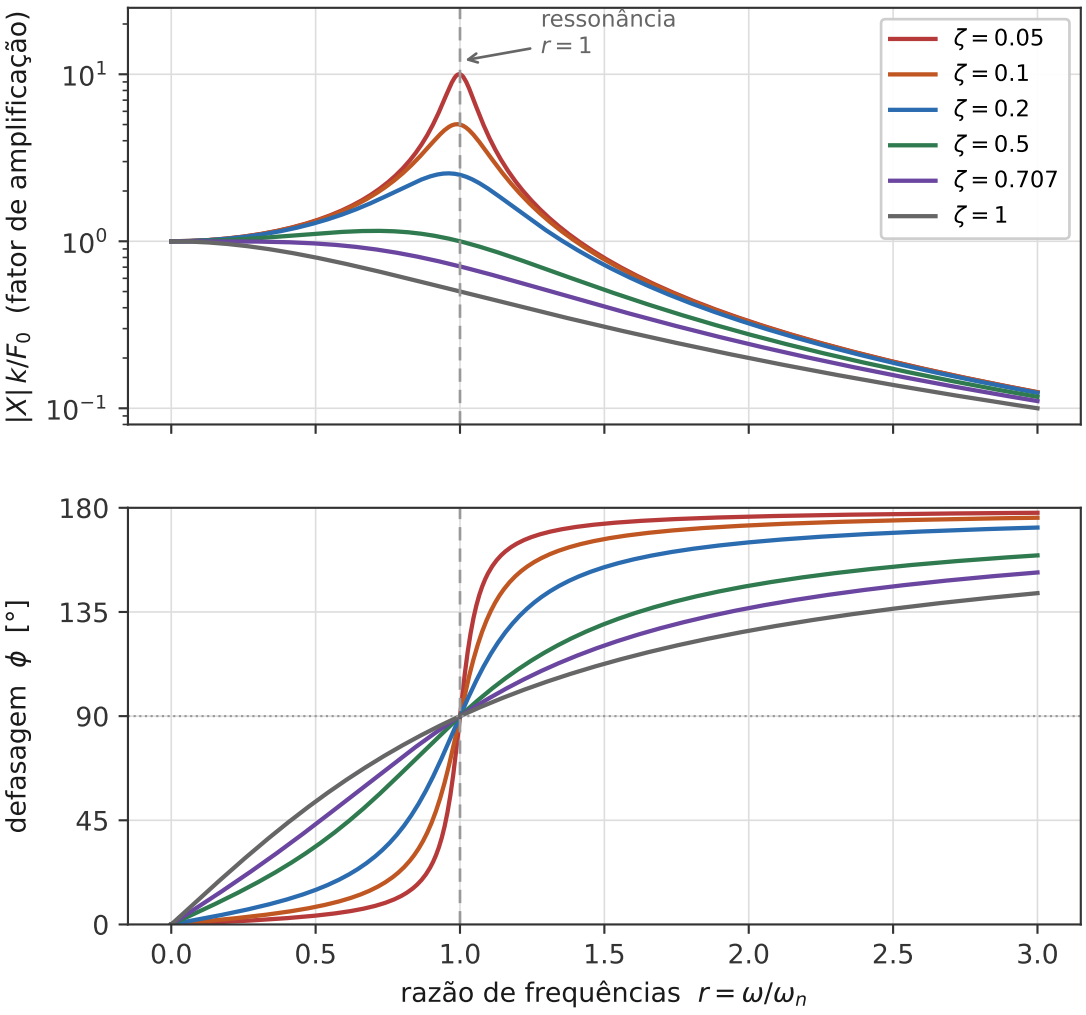
ressonância
$r = 1$
$\zeta = 0.05$
$\zeta = 0.1$
$\zeta = 0.2$
$\zeta = 0.5$
$\zeta = 0.707$
$\zeta = 1$
$|X|\,k/F_0$ (fator de amplificação)
10^1
10^0
10^{-1}
defasagem ϕ [°]
180
135
90
45
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
razão de frequências $r = \omega/\omega_n$